

DATA-DRIVEN INSIGHTS INTO GLOBAL AIR QUALITY

A Statistical Analysis of Environmental and Demographic Factors Affecting AQI



INTRODUCTION

Air quality is a critical global health and environmental concern. This project investigates key factors that may influence the Air Quality Index (AQI) across countries, including greenhouse gas (GHG) emissions, forest cover, and population density.

Hypothesis

- H0: 'There is no correlation between AQI value and GHG emissions.'
- H1: 'There is a positive correlation between AQI value and GHG emissions.'
- H0: 'There is no correlation between AQI value and forest cover.'
- H1: 'There is a negative correlation between AQI value and forest cover.'
- H0: 'There is no correlation between AQI value and population density.'
- H1: 'There is a positive correlation between AQI value and population density.'
- H0: 'There is no difference in mean AQI values among different continents.'
- H1: 'There is a significant difference in mean AQI values among different continents.'

Analysis 1

1) Significant positive correlation between GHG emissions and AQI: Higher GHG emissions are associated with higher AQI values (correlation coefficient of approximately 0.60, $p < 0.05$).

Analysis 2

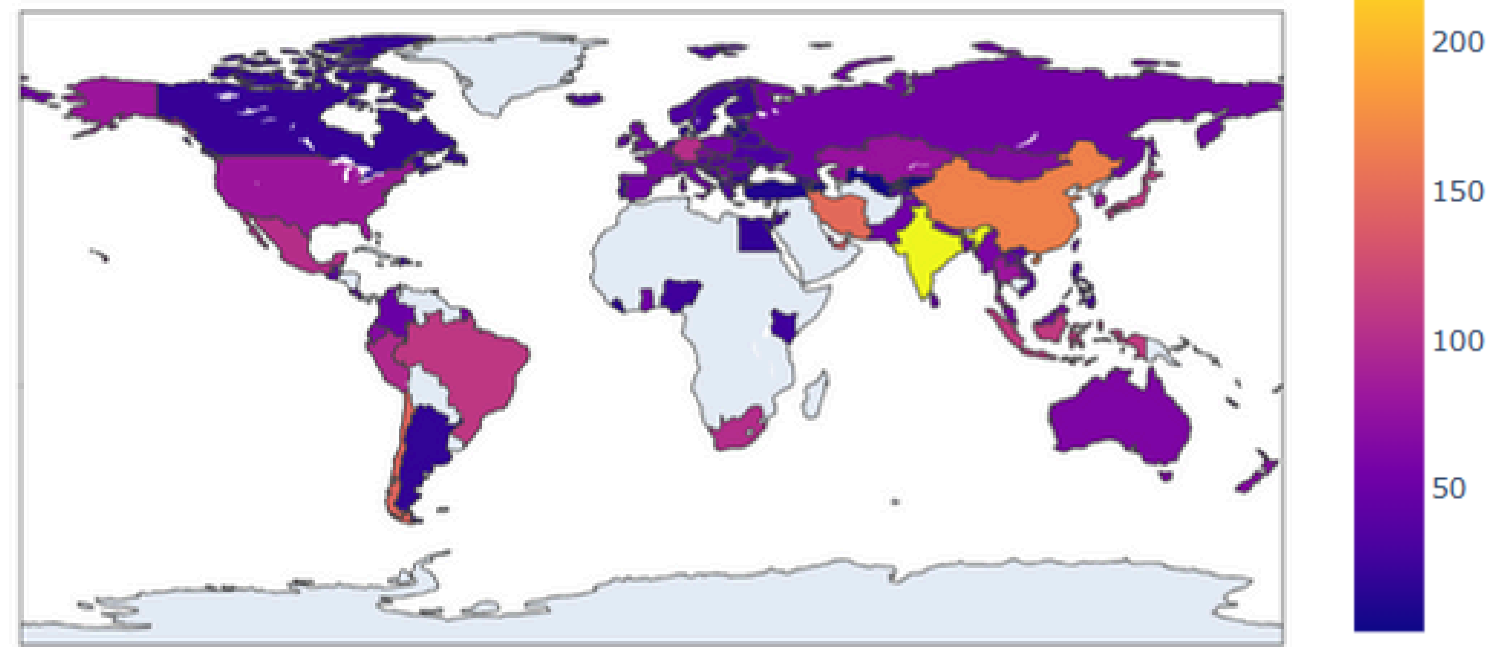
2) No significant correlation between forest cover and AQI: The analysis did not reveal a statistically significant relationship between forest cover and AQI.

Analysis 3

3) No significant correlation between population density and AQI: No statistically significant relationship was observed between population density and AQI.

Analysis 4

4) Significant differences in AQI values across continents: We found significant difference across different continents. $p < 0.05$



CONCLUSION

Our analysis showed a significant positive correlation between greenhouse gas emissions and poorer air quality (higher AQI). However, no strong evidence was found linking forest cover or population density to AQI values. Additionally, air quality levels varied significantly across continents. These findings highlight the global impact of emissions on air quality and suggest that other factors beyond forest cover and population density may play a larger role in shaping pollution levels.

